

A Posterior Predictive for Pass@ k

Per-Prompt Coverage Estimation for Post-Training Diagnostics

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Abstract

A prompt that fails a handful of scored attempts is rated impossible by the extrapolation $1 - (1 - \hat{p})^k$, and rated so at every k . The bias of this plug-in rule has been known since [Chen et al. \(2021\)](#) flagged it, and the population-level repair, a two-parameter difficulty prior fitted by marginal maximum likelihood, has recently been published by [Kazdan et al. \(2025\)](#) for aggregate coverage curves. What remains open, and what post-training diagnostics actually need, is the per-prompt version: the posterior expectation $\mathbb{E}[1 - (1 - p)^k \mid s]$, which conditions on each prompt’s own count. We prove it is the exact conditional success probability under the model, optimal under Brier and log loss, and aggregate-consistent with the published population estimator. On nine thousand prompts with eight scored samples each, the plug-in scores held-out log loss 3.37, driven entirely by zero-success prompts that then succeed; the posterior predictive scores 0.48 and lands the extrapolated aggregate curve within two points where the plug-in is thirteen under. Probit-normal and beta-binomial priors tie, so nothing hangs on the family.

1 Introduction

Language models are improved after pretraining by having them attempt problems and scoring the attempts automatically: a unit test passes, a final answer matches, a verifier accepts. The evaluations that steer this process sample each prompt many times, because a model that solves a problem once in sixteen tries can often be trained into solving it reliably. Pass@ k , the probability that at least one of k sampled attempts succeeds, averaged over prompts, is the working currency of post-training evaluation, and decisions of real consequence ride on it: which prompts to train on, whether a fine-tuning recipe helped, when to stop sampling.

When $n \geq k$ scored attempts per prompt are available, the combinatorial estimator of [Chen et al. \(2021\)](#), $1 - \binom{n-c}{k} / \binom{n}{k}$ with c the correct count, is unbiased and settles the matter. The trouble begins when k exceeds the attempts at hand and the curve must be extrapolated. The common device evaluates the coverage formula at a point estimate: from $\hat{p} = s/m$, report $1 - (1 - \hat{p})^k$. We will call this the plug-in. It is not a straw man. [Malladi et al. \(2026\)](#) write “we convert pass@1 to an estimated pass@16 value using $f_{16}(p) := 1 - (1 - p)^{16}$ ” inside the coverage diagnostic that decides where their filtered fine-tuning method applies; [Barakat et al. \(2026\)](#) build a pass@ k optimization analysis on the same per-prompt functional; and every pass@ k curve reported beyond its sampled depth uses some version of it. [Chen et al. \(2021\)](#) already cautioned that the plug-in is biased, and their combinatorial estimator repairs the interpolation regime $k \leq n$; it offers nothing for extrapolation, which is where the diagnostics live.

This note supplies and prices the per-prompt repair. The plug-in has two defects that a two-parameter model removes: it is biased downward for every prompt difficulty, by convexity, and it assigns probability exactly zero to any prompt with no observed successes, an impossibility verdict that later attempts routinely refute. The repair replaces evaluation at a point estimate with a posterior expectation, $\mathbb{E}[1 - (1 - p)^k \mid s]$, under a difficulty prior fitted once across prompts by marginal maximum likelihood. At the population level this idea has just been published: [Kazdan et al. \(2025\)](#) fit a beta prior by marginal likelihood and report the prior expectation of coverage, obtaining well-calibrated aggregate pass@ k curves. The per-prompt object is the missing piece. Diagnostics act prompt by prompt, the zero-success pathology is a per-prompt pathology, and conditioning on each prompt’s own count is what a posterior is for. We prove the conditional version is the exact per-prompt success probability under the model, optimal under both proper scoring rules in play, and aggregate-consistent with the published population estimator, then measure all of it on released rollout records.

2 Related work

[Brown et al. \(2024\)](#) established repeated sampling as an inference-time scaling axis and model aggregate coverage curves with exponentiated power laws, computing per-problem pass rates by the combinatorial estimator. [Schaeffer et al. \(2025\)](#) explain those power laws by the heavy-tailed distribution of per-problem success probabilities, fitting a three-parameter Kumaraswamy law to plug-in per-problem estimates and forecasting coverage through the same integral that defines aggregate pass@ k ; their per-problem inputs are point estimates, not posteriors. [Kazdan et al. \(2025\)](#) are closest to this note: a beta difficulty prior fitted by marginal maximum likelihood, aggregate coverage reported as the prior expectation of $1 - (1 - p)^k$, with zero-success problems handled by the prior mass rather than declared impossible, and a budget-allocation application. They never condition on an individual prompt’s count; every prompt receives the same curve. The present note is the conditional complement: the posterior predictive per prompt, its optimality, and its measured advantage where diagnostics operate. Elsewhere in the evaluation literature, [Hariri et al. \(2025\)](#) replace pass@ k reporting with Bayesian credible intervals at the model level, and [Miller \(2024\)](#) bring frequentist standard errors to eval reporting; neither extrapolates per-prompt coverage. [Malladi et al. \(2026\)](#) supply the motivating diagnostic, and nothing here disputes their method: the question is confined to the extrapolation step inside it.

3 The estimation problem

The practitioner’s situation is this. There are N prompts and a compute budget that allows m scored attempts at each, so the data are success counts $s_1, \dots, s_N$ with $s_i \in \{0, \dots, m\}$. Decisions are then taken about a larger budget $k > m$ that will not be spent yet, or ever: whether a prompt is worth training on depends on its chance of being solved within k attempts, and whether a fine-tuning method helped depends on how many prompts cross that bar. [Malladi et al. \(2026\)](#) instantiate the situation with $k = 16$ extrapolated from pass@1 estimates; any pass@ k curve reported beyond its sampled depth instantiates it too. The problem is to choose the function of s_i that stands in for the unavailable deeper measurement, and the quality of a choice is measured the way predictions are: per prompt by log loss and Brier score against later attempts, in aggregate by the error of the reported curve.

The model under which the problem has an exact answer is minimal. Each prompt carries a latent per-attempt success probability p_i , drawn independently from a difficulty distribution G on $[0, 1]$; conditional on p_i , attempts are independent Bernoulli(p_i) trials, so $s_i \sim \text{Bin}(m, p_i)$. Two estimands follow. Per prompt, the probability that a fresh batch of k attempts contains a success,

$$q_k(p_i) = 1 - (1 - p_i)^k, \quad (1)$$

to be predicted from s_i alone; and in aggregate, the pass@ k the whole population would exhibit,

$$\pi_k = \int_0^1 (1 - (1 - p)^k) dG(p). \quad (2)$$

The conditional independence of attempts given p_i is not the assumption under attack: attempts are drawn with independent seeds, and the model grants that. What varies across the field is p_i itself, and the whole question is how to respect that variation when s_i is a noisy glimpse of it. Evaluating q_k at s_i/m is one answer; taking the posterior expectation of q_k is the other; the next two sections price them.

4 The plug-in and its failure mode

The plug-in predictor evaluates q_k at the point estimate $\hat{p}_i = s_i/m$:

$$\widehat{\text{pass}}_k^{\text{plug}} = 1 - \left(1 - \frac{s_i}{m}\right)^k. \quad (3)$$

At $s_i = 0$ the prediction is exactly zero at every k , however small m is, so each eventual success on such a prompt costs unbounded log loss. The aggregate defect was flagged by [Chen et al. \(2021\)](#); the following restates it with the strictness and the $k = 1$ boundary that the experiments confirm.

Proposition 1 (The plug-in is biased downward for every difficulty). Fix $m \geq 1$ and $k \geq 2$, and let $s \sim \text{Bin}(m, p)$. Then

$$\mathbb{E} \left[1 - \left(1 - \frac{s}{m}\right)^k \right] \leq 1 - (1 - p)^k, \quad (4)$$

with equality only at $p \in \{0, 1\}$; at $k = 1$ equality holds for all p . Consequently the aggregate plug-in curve lies below π_k for every G not concentrated on $\{0, 1\}$.

Proof. Write $\varphi(x) = (1 - x)^k$ on $[0, 1]$. Its second derivative is $\varphi''(x) = k(k - 1)(1 - x)^{k-2} \geq 0$, strictly positive on $[0, 1)$ when $k \geq 2$, so φ is convex, and strictly convex there. The estimate $\hat{p} = s/m$ satisfies $\mathbb{E} \hat{p} = p$. Jensen's inequality gives $\mathbb{E} \varphi(\hat{p}) \geq \varphi(\mathbb{E} \hat{p}) = (1 - p)^k$, and the inequality is strict whenever $\hat{p}$ is nondegenerate, which is exactly $p \in (0, 1)$. Negating and adding one reverses the inequality into the display. At $k = 1$, φ is affine and Jensen holds with equality. The aggregate statement follows by integrating the pointwise inequality against G . $\square$

The bias grows with the curvature of φ , hence with k , and shrinks with the variance of $\hat{p}$, hence with m : extrapolating further from fewer samples costs more, which is the regime diagnostics live in.

5 The posterior predictive

Give p a prior π_θ with two parameters fitted by marginal maximum likelihood across prompts, and predict with the posterior expectation

$$\widehat{\text{pass}}_k^{\text{post}} = \mathbb{E}[1 - (1 - p)^k \mid s]. \quad (5)$$

Two convenient families serve. Under a Beta(a, b) prior the posterior is Beta($a + s, b + m - s$) and the moment is a falling product,

$$\mathbb{E}[(1 - p)^k \mid s] = \prod_{j=0}^{k-1} \frac{b + m - s + j}{a + b + m + j}, \quad (6)$$

so Equation (5) is closed-form. Under a probit-normal prior, $p = \Phi(\theta)$ with $\theta \sim N(\mu, \tau^2)$, the marginal likelihood of the observed counts and the posterior moment are one-dimensional Gauss–Hermite quadratures over θ ; this is the family that composes with correlated extensions, since a factor-correlated portfolio of successes conditions to exactly this form (Cotton, 2026). In the exchangeable setting of this note the two families are competitors on equal footing, and the study below treats them as such.

Proposition 2 (The posterior predictive is the target). Let Y indicate a success among k fresh attempts at the same prompt, drawn independently of the training attempts given p . Under the model, with prior equal to the true difficulty law G : (i) $\mathbb{E}[Y \mid s] = \mathbb{E}[1 - (1 - p)^k \mid s]$, so the posterior predictive is the exact conditional success probability; (ii) it minimizes expected Brier score and expected log loss among all predictors that depend on the data only through s ; (iii) its average over prompts is exactly π_k .

Proof. Given p , Y is Bernoulli with parameter $q_k(p) = 1 - (1 - p)^k$ and is independent of s . The tower property gives $\mathbb{E}[Y \mid s] = \mathbb{E}[\mathbb{E}[Y \mid p, s] \mid s] = \mathbb{E}[q_k(p) \mid s]$, which is (i). For (ii), the conditional expectation minimizes mean squared error among s -measurable predictors, which is the Brier statement, and the true conditional probability minimizes expected log loss because the excess log loss of any other predictor is the expectation of a Kullback–Leibler divergence, which is nonnegative. For (iii), $\mathbb{E}[\mathbb{E}[q_k(p) \mid s]] = \mathbb{E}[q_k(p)] = \pi_k$, the tower property again. $\square$

Part (iii) locates the published population estimator inside this one: averaging the posterior predictive over prompts recovers the prior expectation of coverage, which is precisely what Kazdan et al. (2025) report. Their estimator is the aggregate shadow of Equation (5); conditioning on s_i is what the posterior adds, and the per-prompt columns of the study below are where the addition shows. Proposition 2 holds at the true G ; in practice G is replaced by a two-parameter family fitted by marginal maximum likelihood, and the study measures what survives that substitution.

6 Numerical study on released rollouts

The data are the publicly released Pass8 rollout records of the RLVE training suite: procedurally generated reasoning tasks, each with an automatic verifier, of the kind used as reinforcement-learning environments for language models. They carry nine thousand prompts across eighteen environment families, eight scored samples per prompt from the Qwen3-4B-Thinking model, three

gigabytes of response text reduced to a forty-one-kilobyte outcome table.¹ Samples 0–3 of each prompt are the training half; samples 4–7 are held out. The held-out per-prompt pass@4 is then a single Bernoulli outcome, whether any of the four succeeded, so predicted probabilities are scored by log loss and Brier score with no estimator ambiguity. The fitted priors are Beta(0.40, 0.83) and probit-normal $\mu = -0.78$, $\tau = 1.39$; both put substantial mass near $p = 0$ and $p = 1$, which is what the environments look like.

predictor of held-out pass@4	log loss	Brier
plug-in $1 - (1 - s/4)^4$	3.367	0.192
posterior predictive, probit-normal	0.483	0.160
posterior predictive, beta-binomial	0.483	0.160

Table 1: Per-prompt scores over nine thousand held-out Bernoulli outcomes. The plug-in’s log loss is dominated by zero-training- success prompts that succeed in the held-out half; the posterior predictive prices those prompts at their shrunk posterior rate instead of zero. The two prior families tie.

aggregate	$k = 1$	$k = 2$	$k = 4$	$k = 8$
truth (all eight samples)	0.322	0.442	0.569	0.682
plug-in from the training half	0.322	0.412	0.495	0.549
probit-normal posterior	0.323	0.444	0.561	0.663
beta-binomial posterior	0.323	0.444	0.560	0.661

Table 2: Aggregate pass@ k extrapolated from four samples per prompt, against the estimator of [Chen et al. \(2021\)](#) on all eight. The plug-in’s downward bias grows with k as Proposition 1 requires, reaching thirteen points absolute at $k = 8$; the posterior predictives land within two points. By Proposition 2(iii) these aggregate rows also measure the population estimator of [Kazdan et al. \(2025\)](#) under the same fitted priors; the per-prompt scores of Table 1 are what conditioning adds beyond it.

The per-prompt comparison in Table 1 is seven-fold in log loss and twenty percent in Brier, for two fitted parameters. The aggregate comparison in Table 2 shows the plug-in bias growing with the extrapolation distance exactly as the convexity requires.

7 Scope

The population-level repair is prior art: [Kazdan et al. \(2025\)](#) fit the same beta family the same way and report well-calibrated aggregate curves, and [Schaeffer et al. \(2025\)](#) model the difficulty distribution with a richer family for the same integral. The contribution here is confined to the conditional object: the per-prompt posterior predictive, its optimality under the scoring rules diagnostics use, its aggregate consistency with the published estimator, and the measured seven-fold per-prompt gap. The plug-in bias itself has been on record since [Chen et al. \(2021\)](#); Proposition 1 adds strictness, the $k = 1$ boundary, and the aggregate consequence.

¹Dataset: CL-From-Nothing/RLVE-Qwen3-4B-Thinking-2507-Pass8-Rollouts on Hugging Face. Extraction and experiment scripts are committed under `research/cavity_calculus/exp2_passk/` in the `winning` repository.

The bias scales with the noise in $\hat{p}$, hence shrinks as the per-prompt sample count grows; the study uses four training samples per prompt, and [Malladi et al. \(2026\)](#) do not state the count behind their pass@1 estimates, so the numbers here bound the mechanism, not the magnitude in their setting. The data cover one model on one task suite; transfer to standard math and code benchmarks is plausible and unshown. The two prior families are statistically indistinguishable here, so the choice between them can be made on convenience or, when correlated portfolios of attempts arise, on composability ([Cotton, 2026](#)). Diagnostics built on the plug-in, including the coverage ratio of [Malladi et al. \(2026\)](#), accept the posterior predictive as a drop-in for their extrapolation step, and measuring that substitution in their setting is the natural next experiment.

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